

INTERNSHIP PROJECT REPORT TASK-6

Title

Design and implement a custom SIEM dashboard (using ELK or Security Onion) that visualizes critical security events like brute-force attacks, privilege escalations or exfiltration attempts.

1. Introduction

As part of my cybersecurity internship, I worked on a project to create a custom SIEM dashboard using Elastic Security and Kibana. The main objective of this task was to visualize important security events and monitor suspicious activities through a centralized dashboard.

During this project, I created different visualizations to display brute-force attacks, privilege escalation events, and data exfiltration activities. This task helped me understand how SIEM tools are used in Security Operations Centers (SOC) to monitor security events, detect threats and improve overall cybersecurity awareness.

2. Background

Security Information and Event Management (SIEM) is an important technology used by organizations to collect, analyze, and monitor security logs from different systems. It helps security teams identify suspicious activities and respond to security incidents more effectively.

For this task, Elastic Security and Kibana were used to create a custom SIEM dashboard. Sample security events such as brute-force attacks, privilege escalation attempts, and data exfiltration activities were generated and visualized. The dashboard provides a clear view of security events, making it easier to monitor threats and understand security trends.

3. Learning Objectives

The main objective of this task was to understand how a SIEM platform is used to monitor and analyze security events. It was also intended to provide practical experience in working with Elastic Security and Kibana for security monitoring.

The main objectives of this project were:

- To understand the basics of SIEM and security monitoring.
- To create custom visualizations and dashboards for security monitoring.
- To learn how to identify brute-force attacks, privilege escalation events, and data exfiltration activities.
- To understand how security logs are collected and analyzed.
- To gain practical experience in log analysis and threat detection.

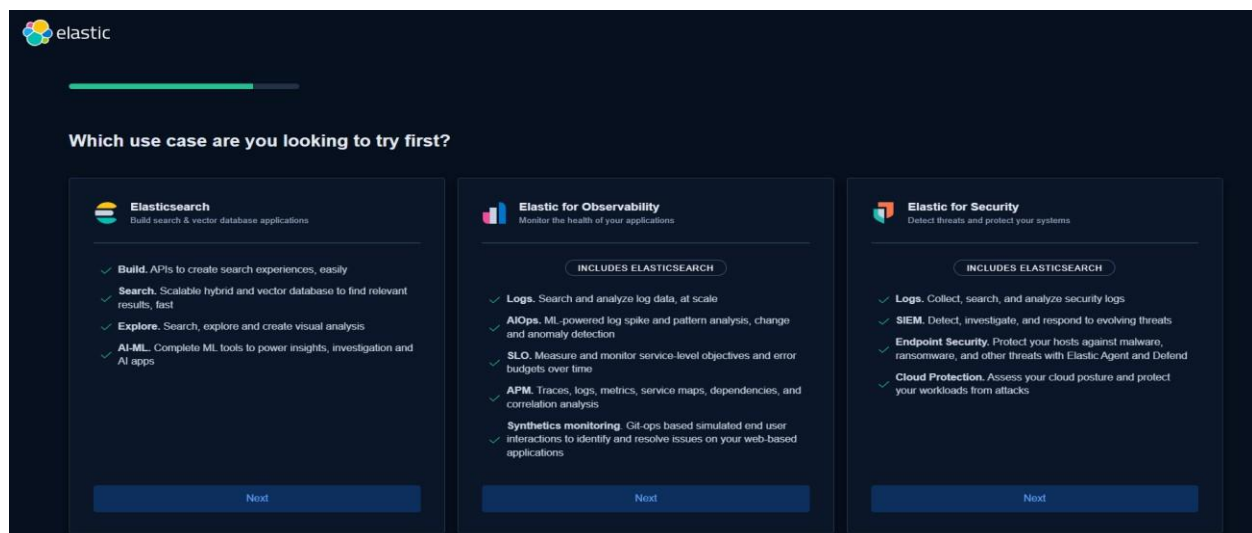
4. Activities and Tasks Performed

Step 1 - Creating Elastic Cloud Environment

I started the project by creating an Elastic Cloud account and selecting Elastic Security during the onboarding process. This helped me understand how SIEM platforms can be used for security monitoring and threat detection.

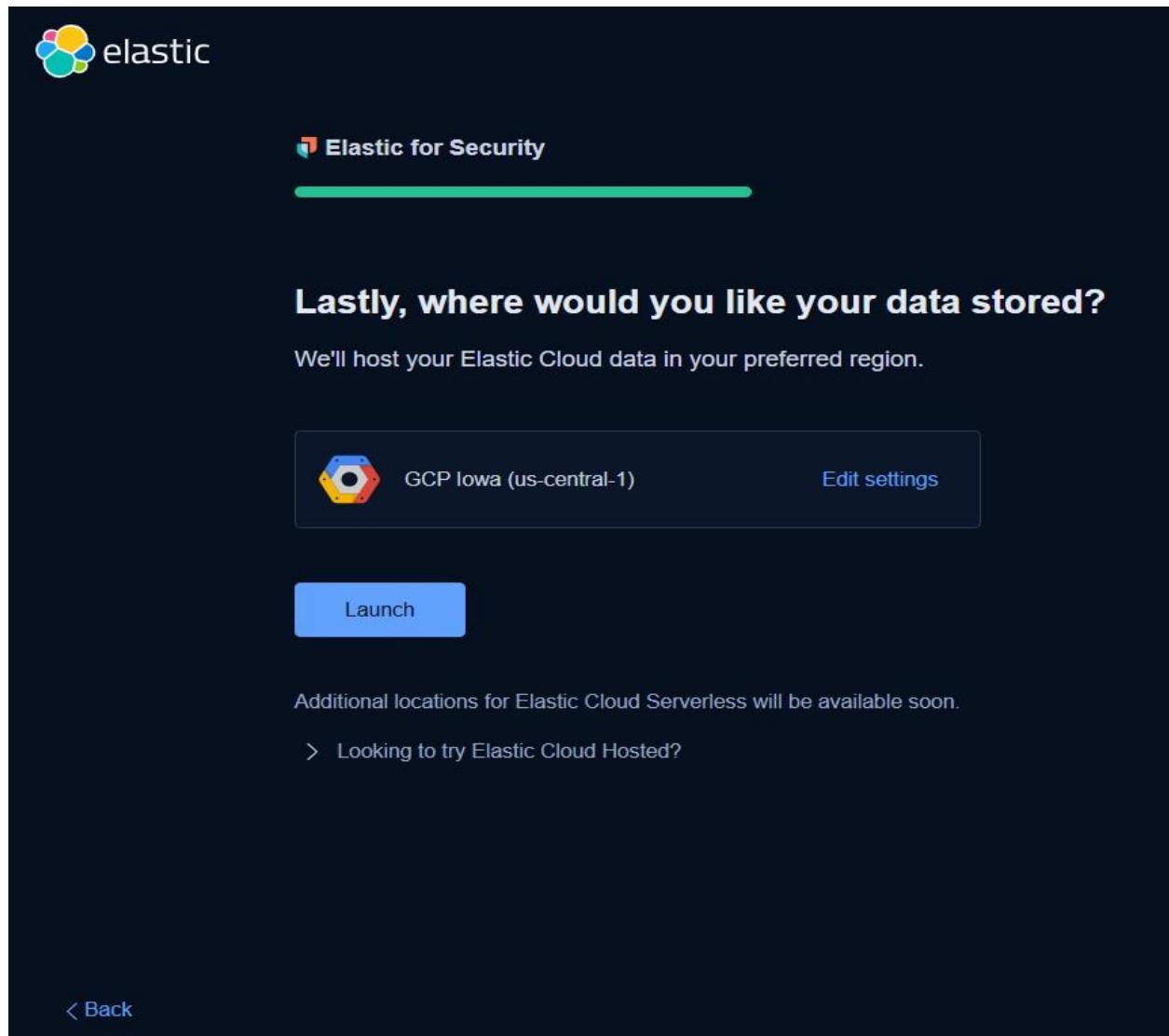
Step 2 - Elastic Security Use Case Selection

During the onboarding process, Elastic Cloud provided multiple use case options. For this project, I selected Elastic Security because it provides SIEM capabilities, security monitoring, threat detection, and alert management required for developing and testing custom detection rules.



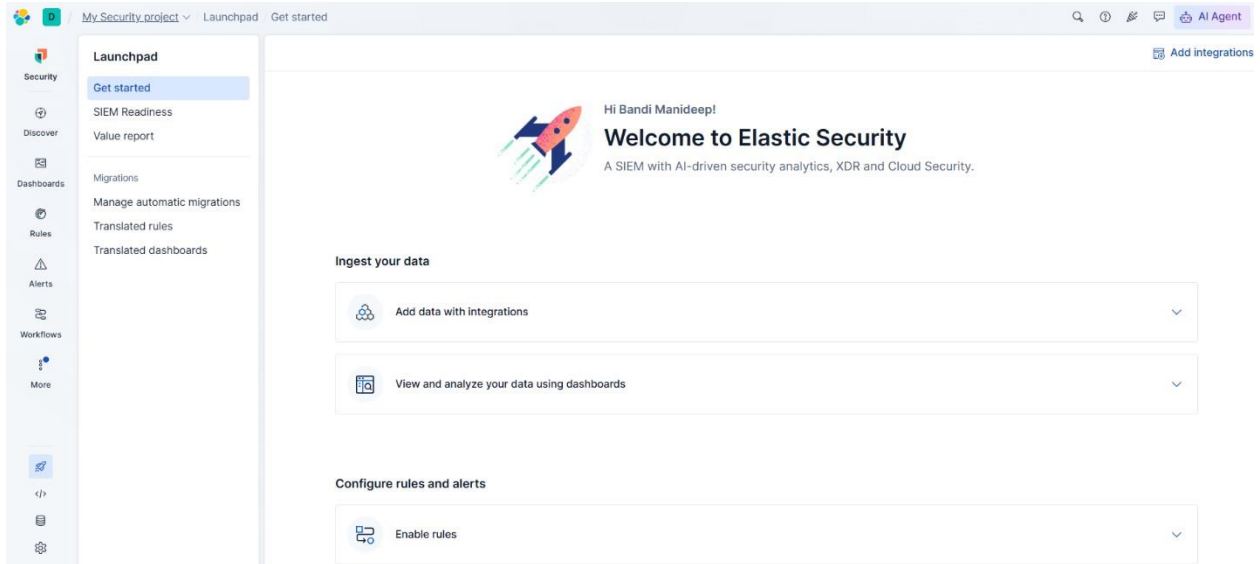
Step 3 - Selecting the Cloud Region and Launching the Project

In this step, I selected the cloud region where the Elastic Security project data would be stored. After reviewing the available settings, I kept the default region and launched the Elastic Security environment. This created the cloud-based SIEM workspace that would be used for log analysis, visualization creation and dashboard development.



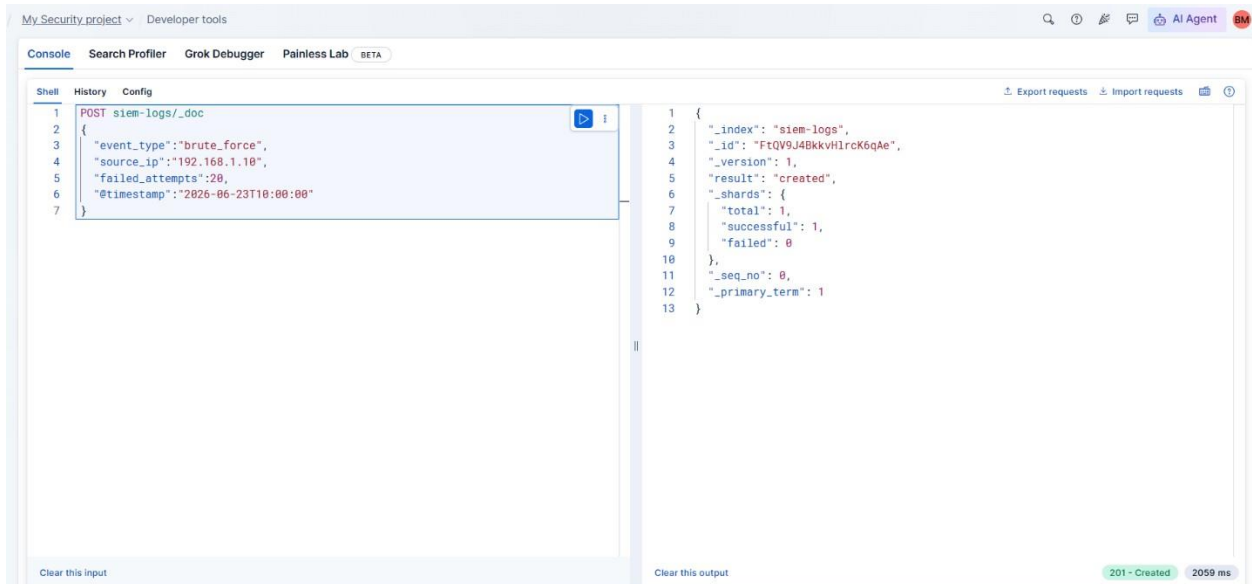
Step 4 - Accessing Elastic Security Dashboard

After completing the deployment setup, the Elastic Security dashboard was successfully accessed. I explored the available features such as dashboards, rules, alerts, and data analysis options. This environment served as the central platform for creating visualizations, managing security data and building the custom SIEM dashboard required for the project.



Step 5 - Creating and Storing a Brute-Force Attack Log

In this step, I used the **Developer Tools Console** in Elastic Security to create a sample brute-force attack log. The log contained information such as the event type (**brute_force**), source IP address (**192.168.1.10**), number of failed login attempts (**20**), and the event timestamp. After executing the query, Elastic successfully created the document in the **siem-logs** index and returned a **201 Created** response, confirming that the security event was stored successfully. This log was later used for creating visualizations and building the SIEM dashboard.



The screenshot displays the Elastic Security Developer Tools Console. The left pane shows the input JSON document for a brute-force attack log. The right pane shows the resulting JSON response from the Elasticsearch API, indicating a successful creation of the document.

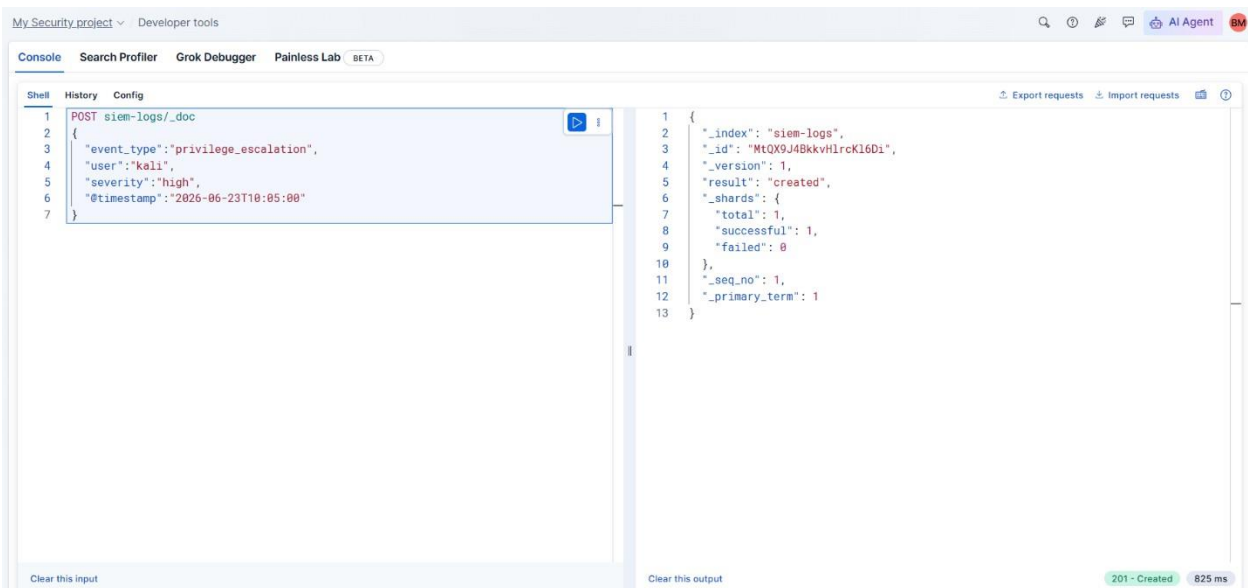
```
1 POST siem-logs/_doc
2 {
3   "event_type": "brute_force",
4   "source_ip": "192.168.1.10",
5   "failed_attempts": 20,
6   "@timestamp": "2026-06-23T10:00:00"
7 }
```

```
1 {
2   "_index": "siem-logs",
3   "_id": "FtQV9J4BkkvH1rcK6qAe",
4   "_version": 1,
5   "result": "created",
6   "_shards": {
7     "total": 1,
8     "successful": 1,
9     "failed": 0
10  },
11   "_seq_no": 0,
12   "_primary_term": 1
13 }
```

201 - Created 2059 ms

Step 6 - Creating and Storing a Privilege Escalation Event Log

In this step, I created a sample **privilege escalation** security event using the Developer Tools Console. The log included details such as the event type (**privilege_escalation**), username (**kali**), severity level (**high**), and event timestamp. After executing the query, Elastic successfully stored the log in the **siem-logs** index and returned a **201 Created** response. This event was later used to visualize privilege escalation activities within the SIEM dashboard.



The screenshot displays the Elastic Developer Tools Console interface. The top navigation bar includes 'My Security project' and 'Developer tools'. Below this, tabs for 'Console', 'Search Profiler', 'Grok Debugger', and 'Painless Lab' are visible, with 'Console' being the active tab. The 'Shell' section on the left shows a REST client request:

```
1 POST siem-logs/_doc
2 {
3   "event_type": "privilege_escalation",
4   "user": "kali",
5   "severity": "high",
6   "@timestamp": "2026-06-23T10:05:00"
7 }
```

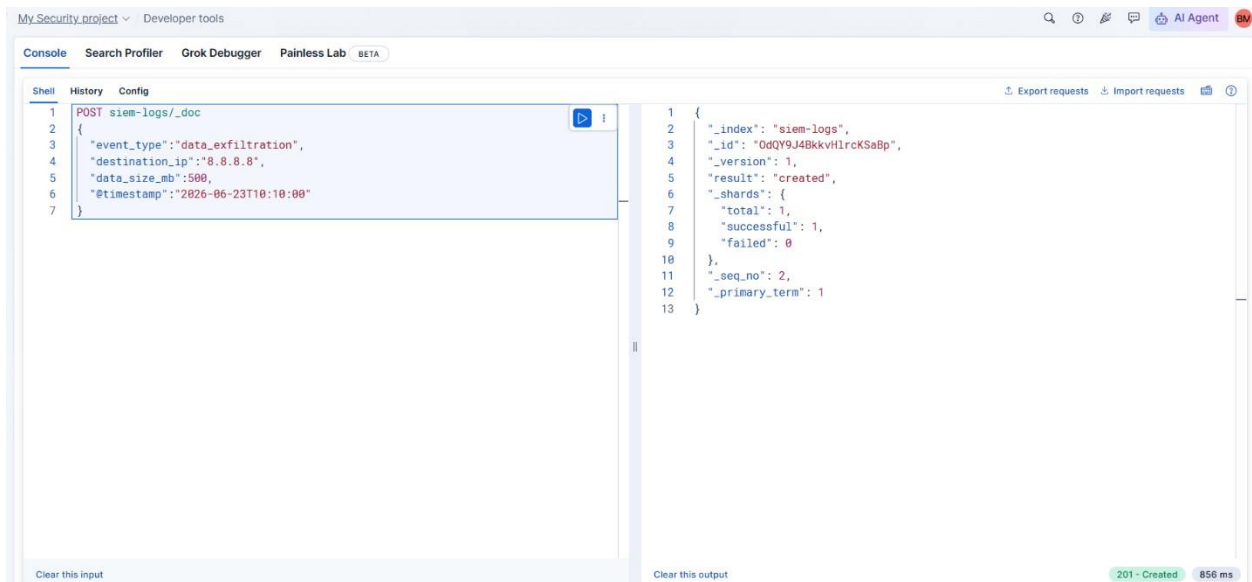
The 'Response' section on the right shows the JSON response from the server:

```
1 {
2   "_index": "siem-logs",
3   "_id": "MtQX9J4BkkvH1rcK16D1",
4   "_version": 1,
5   "result": "created",
6   "_shards": {
7     "total": 1,
8     "successful": 1,
9     "failed": 0
10  },
11   "_seq_no": 1,
12   "_primary_term": 1
13 }
```

At the bottom right, a status bar indicates the response was '201 - Created' and took '825 ms'.

Step 7 - Creating and Storing a Data Exfiltration Event Log

In this step, I created a sample **data exfiltration** event using the Elastic Developer Tools Console. The log contained information such as the event type (**data_exfiltration**), destination IP address (**8.8.8.8**), transferred data size (**500 MB**), and timestamp. After executing the request, Elastic successfully stored the event in the **siem-logs** index and returned a **201 Created** status, confirming that the log was indexed successfully. This event was later used to visualize data exfiltration activity in the SIEM dashboard.



The screenshot displays the Elastic Developer Tools Console interface. The top navigation bar includes 'My Security project', 'Developer tools', and various utility icons. Below the navigation bar, tabs for 'Console', 'Search Profiler', 'Grok Debugger', and 'Painless Lab' are visible. The 'Console' tab is active, showing a 'Shell' view with a command history and a 'Config' view. The command history shows a POST request to 'siem-logs/_doc' with a JSON body containing event details. The 'Config' view shows the response body, which is a JSON object indicating the document was successfully created. At the bottom right, a status bar shows '201 - Created' and '856 ms'.

```
1 POST siem-logs/_doc
2 {
3   "event_type": "data_exfiltration",
4   "destination_ip": "8.8.8.8",
5   "data_size_mb": 500,
6   "@timestamp": "2026-06-23T10:10:00"
7 }

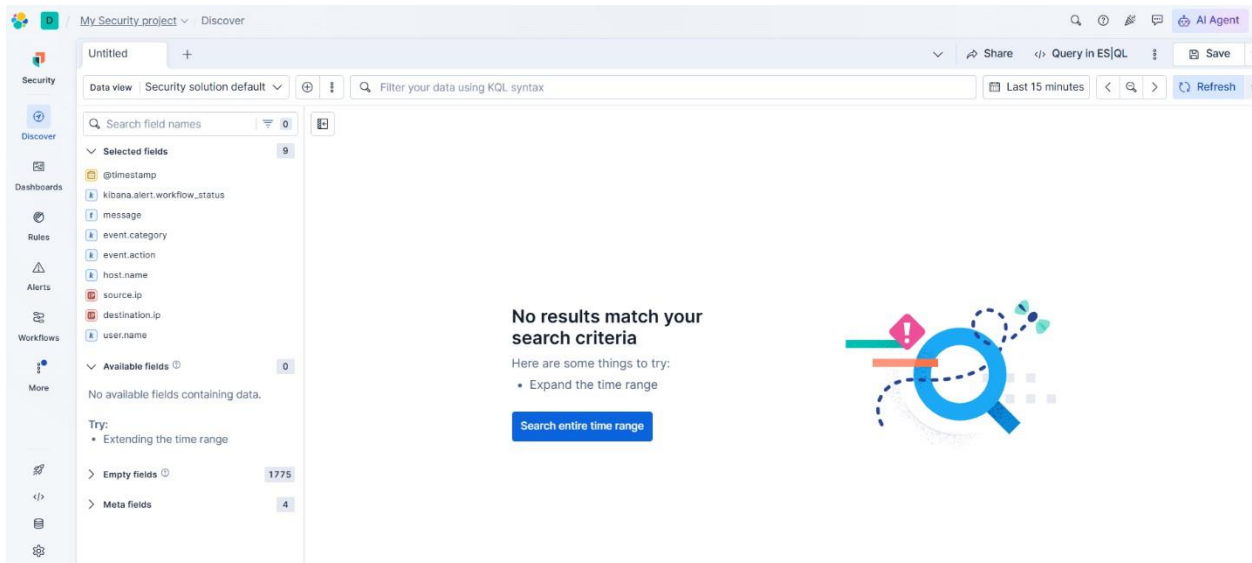
1 {
2   "_index": "siem-logs",
3   "_id": "OdQY9J4BkkvH1rcK5aBp",
4   "_version": 1,
5   "result": "created",
6   "_shards": {
7     "total": 1,
8     "successful": 1,
9     "failed": 0
10  },
11   "_seq_no": 2,
12   "_primary_term": 1
13 }
```

201 - Created 856 ms

Step 8 - Verifying Data in Discover

After creating the sample security events in the **Developer Tools Console**, I navigated to the **Discover** page to verify whether the logs were available for analysis. The Discover section is used to search, filter, and inspect indexed data stored in Elasticsearch. I selected the default security data view and checked the available fields related to security events.

At this stage, no records were displayed because the selected time range was limited to the last 15 minutes, while the generated logs contained different timestamps. This helped me understand how time filters affect data visibility in Elastic Security.



Step 9 - Reviewing Available Data Views

In this step, I opened the **Data Views** section in Elastic to manage the data sources used for log analysis. Data views help Elastic identify and access the indices that store security and system logs. I reviewed the available data views, including the default security-related views, and verified that the required data sources were available for searching and visualization.

This step ensured that the correct data views were configured before performing log analysis and creating dashboard visualizations. Proper data view management allowed Elastic Security to retrieve and display the indexed security events successfully.

The screenshot shows the Elastic Security interface with the 'Data Views' section selected in the left sidebar. The main panel displays a list of data views under the heading 'Data Views'. A search bar is at the top of the list. The table below shows the following data views:

Name	Spaces	Actions
Security solution default (Security Solution Default)	1	Edit, Delete
.kibana-event-log-* (Managed)	1	Edit, Delete
Security solution alerts (Security Solution Managed)	1	Edit, Delete
Security solution attacks (Security Solution Managed)	1	Edit, Delete
logs-* (Managed)	1	Edit, Delete
metrics-* (Managed)	1	Edit, Delete

At the bottom of the table, it indicates 'Rows per page: 10' and a pagination control showing '< 1 >'.

Step 10 - Creating a Custom Data View for Security Logs

In this step, I created a new **Data View** named **siem-logs** to access and analyze the security events stored in Elasticsearch. I configured the index pattern as **siem-logs*** and selected **@timestamp** as the time field. Elastic successfully detected the matching index source, confirming that the data view was connected to the stored logs.

Creating this data view made it easier to search, filter, and visualize the security events generated during the project. This data view was later used in the Discover page and dashboard creation process to analyze brute-force, privilege escalation, and data exfiltration events.

The screenshot shows the Kibana 'Create data view' dialog. On the left is a sidebar with 'Data Views' and a list of existing views. The main area is titled 'Create data view' and contains the following fields:

- Name:** siem-logs
- Index pattern:** siem-logs* (with a help icon)
- Timestamp field:** @timestamp (selected from a dropdown)

Below the timestamp field, there is a note: 'Select a timestamp field for use with the global time filter.' and a link 'Show advanced settings'.

On the right side of the dialog, a green success message states: '✓ Your index pattern matches 1 source.' Below this, there are two tabs: 'All sources' and 'Matching sources'. The 'Matching sources' tab is active, showing a table with one entry:

siem-logs	Index

At the bottom of the dialog, there are two buttons: 'Close' and 'Save data view to Kibana'.

Step 11 - Verifying Security Logs in Discover

In this step, I opened the **Discover** page using the custom **siem-logs** data view to verify that the generated security events were successfully indexed. After selecting the correct data view and time range, the logs became visible in the document list.

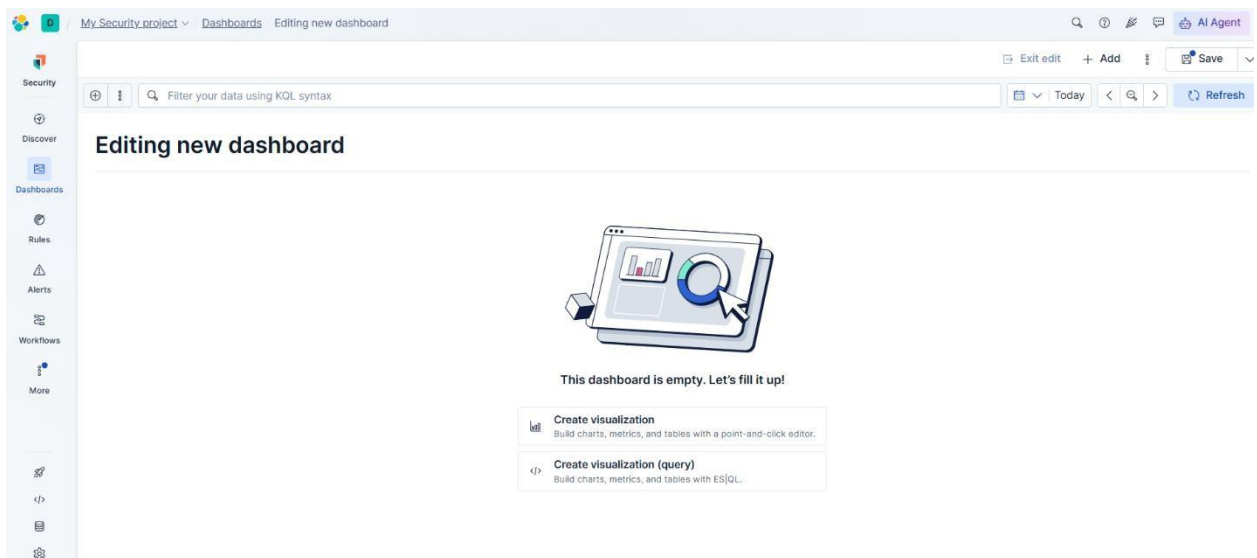
The Discover page displayed the three sample security events created during the project: **Brute Force Attack**, **Privilege Escalation**, and **Data Exfiltration**. I reviewed the event details such as source IP address, user information, severity level, destination IP address, data size and timestamps. This verification confirmed that the logs were successfully stored and could be used for creating SIEM dashboard visualizations.

The screenshot shows the Elasticsearch Discover interface for a project named "My Security project". The "Data view" is set to "siem-logs". The time range is "Last 1 year". The left sidebar shows the "Selected fields" list with "@timestamp" and "data_size_mb". The "Available fields" list includes "@timestamp", "data_size_mb", "destination_ip", "event_type", "failed_attempts", "severity", "source_ip", and "user". The main panel displays a table of documents with 3 rows. The first row shows a data exfiltration event. The second row shows a privilege escalation event. The third row shows a brute force failed attempt event.

Document	Timestamp	Event Type	Severity	User	Source IP	Destination IP	Data Size (MB)	Failed Attempts
1	Jun 23, 2026 @ 15:40:00.000	data_exfiltration	high	kali	192.168.1.10	8.8.8.8	500	20
2	Jun 23, 2026 @ 15:35:00.000	privilege_escalation	high	kali	192.168.1.10	8.8.8.8	500	20
3	Jun 23, 2026 @ 15:30:00.000	brute_force_failed_attempts	high	kali	192.168.1.10	8.8.8.8	500	20

Step 12 - Creating a New SIEM Dashboard

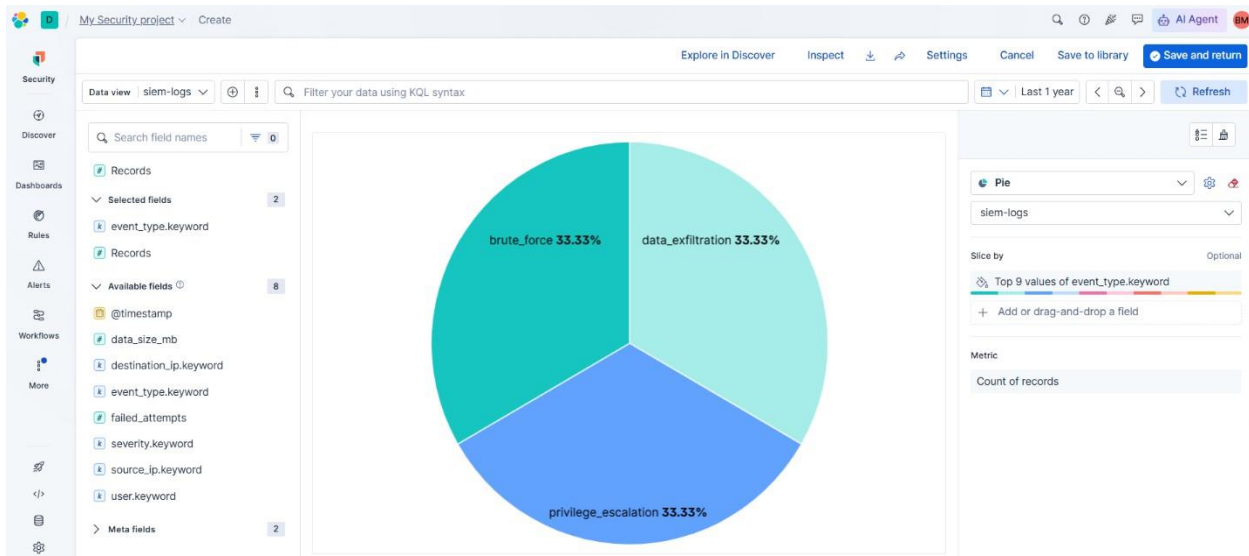
In this step, I navigated to the **Dashboards** section and created a new dashboard for visualizing security events. The dashboard initially appeared empty and provided options to create visualizations using charts, metrics and other graphical components. This served as the starting point for building a custom SIEM dashboard. The purpose of this dashboard was to display important security information such as brute-force attacks, privilege escalation activities and data exfiltration events.



Step 13 - Creating a Security Event Distribution Visualization

In this step, I created a **Pie Chart visualization** using the **siem-logs** data view. The chart was configured to group records based on the **event_type.keyword** field and count the number of events for each category. This visualization provided a clear overview of the different security events stored in the Elasticsearch index.

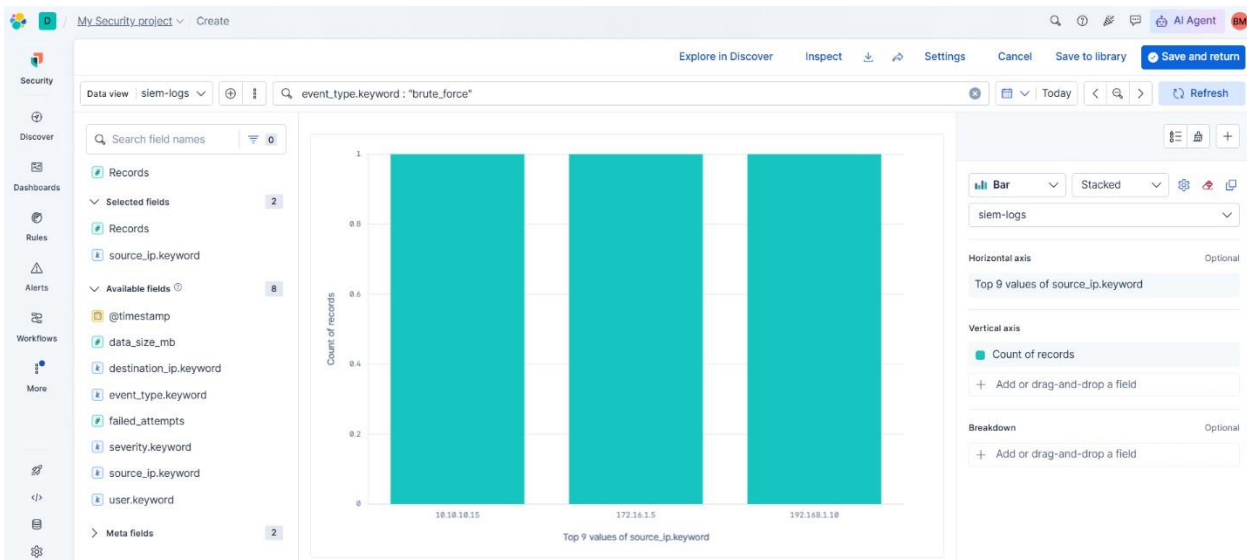
The pie chart displayed the distribution of **Brute Force**, **Privilege Escalation**, and **Data Exfiltration** events. By visualizing the event categories, it became easier to understand the proportion of different security activities and understand how different security events are distributed across the dashboard. This visualization was later added to the SIEM dashboard for monitoring and analysis purposes.



Step 14 -Creating a Brute Force Attack Source IP Visualization

In this step, I created a **Bar Chart visualization** to display the source IP addresses associated with brute-force attack events. I applied a filter using **event_type.keyword : "brute_force"** so that only brute-force attack logs were included in the visualization. The chart was configured to count records and group them based on the **source_ip.keyword** field.

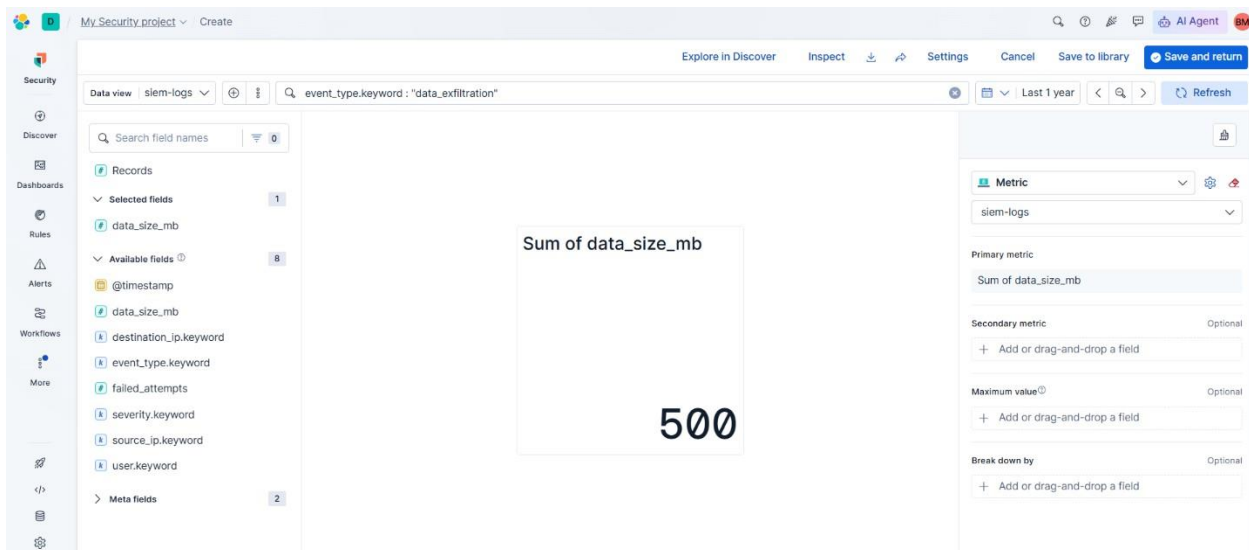
This visualization helped identify the IP addresses involved in brute-force activities and provided a simple way to monitor suspicious login attempts. The bar chart was then added to the SIEM dashboard to improve visibility into attack sources and help identify suspicious source IP addresses.



Step 15 - Creating a Data Exfiltration Metric Visualization

In this step, I created a **Metric visualization** to display the total amount of data involved in data exfiltration events. I applied the filter **event_type.keyword : "data_exfiltration"** to focus only on data exfiltration logs. The **data_size_mb** field was selected as the primary metric, and the aggregation type was set to **Sum**.

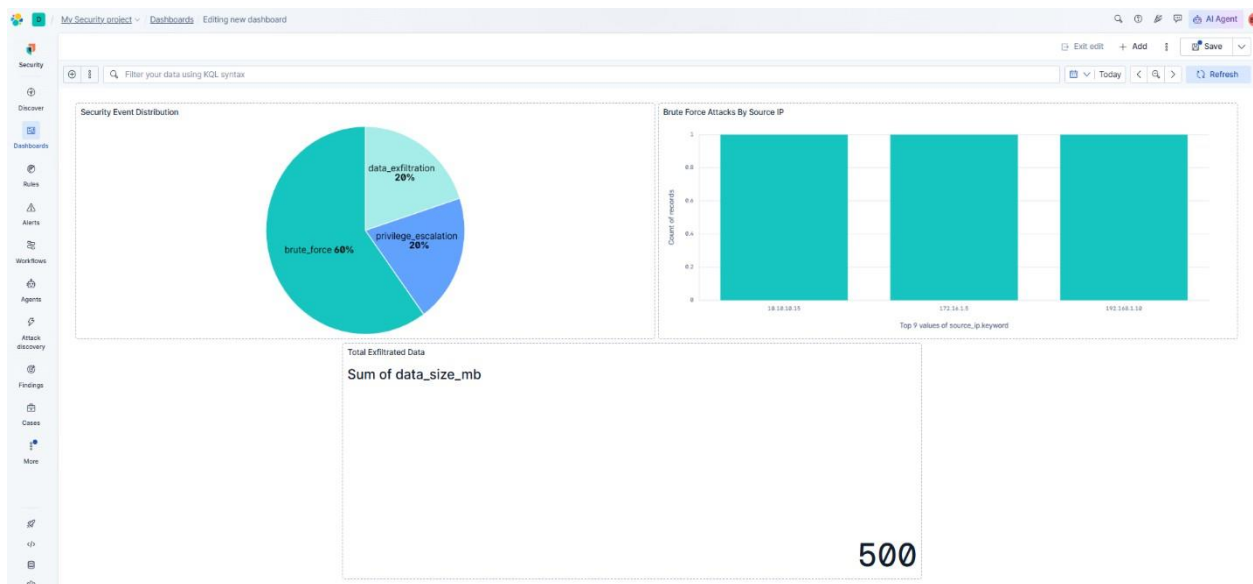
The visualization displayed a total value of **500 MB**, representing the amount of data transferred during the simulated exfiltration activity. This metric was added to the SIEM dashboard to provide a quick overview of potential data leakage and help monitor exfiltration incidents.



Step 16 - Final SIEM Dashboard Creation

In this step, I combined all the created visualizations into a single SIEM dashboard. The dashboard includes a **Security Event Distribution Pie Chart**, a **Brute Force Attacks by Source IP Bar Chart**, and a **Total Exfiltrated Data Metric**. These visualizations were arranged on the dashboard to provide a clear view of the simulated security events.

The completed dashboard helps in monitoring and analyzing different types of security activities from one location. After verifying that all visualizations were displayed correctly, the dashboard was saved successfully for future security analysis and monitoring.



5. Skills and Competencies Gained

- Gained practical experience in using Elastic Security and Kibana for security monitoring.
- Learned how to create and manage data views for analyzing security logs.
- Improved understanding of SIEM concepts and security event management.
- Learned how to generate, store, and verify security event logs in Elasticsearch.
- Improved log analysis skills by reviewing and validating security event data.
- Gained experience in creating different visualizations such as pie charts, bar charts, and metric panels.
- Developed a better understanding of how security teams use SIEM dashboards to monitor and investigate potential threats.

6. Feedback and Evidence

Feedback

This project provided me with practical experience in working with a Security Information and Event Management (SIEM) platform using Elastic Security and Kibana. Through the process of creating security logs, verifying data, building visualizations and designing a custom dashboard, I gained a better understanding of how security events are monitored and analyzed. The hands-on implementation helped me improve my knowledge of security monitoring, log analysis, dashboard creation and threat detection.

Evidence Collected

- Elastic Security deployment and project setup screenshots.
- Developer Tools Console screenshots used for creating security event logs.
- Screenshots showing successful insertion of brute-force, privilege escalation, and data exfiltration logs.
- Data View creation and configuration screenshots.
- Discover page screenshots showing indexed security events.
- Security Event Distribution Pie Chart screenshots.
- Brute Force Attacks by Source IP Bar Chart screenshots.
- Total Exfiltrated Data Metric visualization screenshots.
- Final Custom SIEM Dashboard screenshots.
- Screenshots demonstrating dashboard creation and visualization management activities performed during the project.

7. Challenges and Solutions

During this project, I faced several challenges while working with Elastic Security for the first time. Understanding the dashboard creation process and navigating different sections such as Discover, Data Views, Dashboards, and Visualizations required additional time and practice. To overcome this challenge, I explored each feature step by step and learned its purpose through hands-on use.

Another challenge was that the security logs were not initially visible in the Discover page. After investigating the issue, I found that the selected time range and data view settings were preventing the logs from being displayed. I resolved this problem by creating a custom data view and selecting the correct time range.

I also faced difficulties while creating visualizations because some fields were not displayed as expected. To solve this issue, I verified the indexed data, selected the correct fields and adjusted the visualization settings accordingly.

Although these challenges required additional effort, they helped me improve my understanding of Elastic Security, dashboard creation, log analysis and troubleshooting techniques. The experience also increased my confidence in working with SIEM platforms and security monitoring tools.

8. Outcomes and Impact

This project provided me with valuable hands-on experience in working with a Security Information and Event Management (SIEM) platform. Before starting the project, I had only a basic understanding of SIEM dashboards and security monitoring. Through the practical activities performed during this project, I gained a better understanding of how security events are collected, stored, analyzed and visualized within a centralized platform.

I learned how different security activities such as brute-force attacks, privilege escalation events, and data exfiltration attempts can be monitored through a dashboard. This helped me understand how security analysts use dashboards to identify and investigate potential threats.

The outcome of this project was the improvement of my dashboard creation, log analysis and problem-solving skills. Working with Elastic Security and Kibana also improved my confidence in using SIEM tools and strengthened my practical knowledge of cybersecurity monitoring and threat detection.

9. Conclusion

This project provided me with an opportunity to gain practical experience in creating and using a SIEM dashboard with Elastic Security and Kibana. Throughout the project, I successfully created security event logs, verified data, built different visualizations and combined them into a custom dashboard for security monitoring.

By completing this task, I improved my understanding of SIEM concepts, log analysis, dashboard creation, and security event monitoring. I also gained hands-on experience in visualizing brute-force attacks, privilege escalation events and data exfiltration activities.

Overall, this project was a valuable learning experience that enhanced my cybersecurity knowledge and practical skills. The experience gained from this task will support my future learning and help me build a stronger foundation in the field of cybersecurity.